

STUDENT PERFORMANCE & PLACEMENT PREDICTOR

AI & Machine Learning Mini Project — one-page summary | B.Tech Computer Science & Engineering

Objective. Estimate a student's **academic performance** (Poor–Excellent), **placement readiness** (Not Ready–Highly Ready) and **placement probability** (0–100 %) from 14 academic, skill and experience inputs, and explain which factors raised or lowered each estimate. Delivered as a Streamlit web app with trained scikit-learn/XGBoost pipelines and SHAP explanations.

1,200 students (synthetic)	14 model features	5 × 3 models compared	86.7 % readiness accuracy	7.07 pts probability mean error	18/18 tests passed
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Method

Data cleaning removes 15 duplicate rows and turns 6 impossible entries (e.g. CGPA 12.4) into missing values. Imputation, scaling, one-hot encoding and variance filtering sit inside a scikit-learn *Pipeline*, so they are re-fitted in every cross-validation fold — no data leakage. An 80/20 stratified split (seed 42) gives 960 training and 240 test records. Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and XGBoost were compared for each task; the winner was chosen by 5-fold CV macro-F1 (classification) or RMSE (regression), and the test set was used once.

Results on the held-out test set

Prediction	Selected model	Test result
Academic performance	XGBoost	Accuracy 0.800 macro-F1 0.812
Placement readiness	Logistic Regression	Accuracy 0.867 macro-F1 0.835
Placement probability	XGBoost	MAE 7.07 pts R ² 0.930

Placement readiness — test set, 86.7 % accurate

Actual	Not Ready	80 95%	4 5%	0 0%	0 0%
	Developing	11 24%	31 69%	3 7%	0 0%
	Ready	0 0%	8 22%	27 73%	2 5%
	Highly Ready	0 0%	0 0%	4 5%	70 95%
	Predicted				
	Not Ready	Developing	Ready	Highly Ready	

Almost every error is between neighbouring categories.

Dataset (synthetic, 1,200 students)

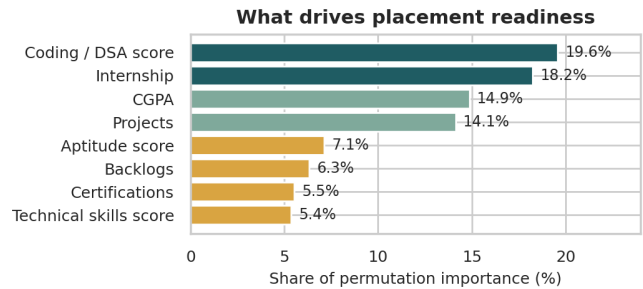
Readiness	Students	Academic	Students
Not Ready	420 (35 %)	Poor	338 (28 %)
Developing	228 (19 %)	Average	370 (31 %)
Ready	184 (15 %)	Good	316 (26 %)
Highly Ready	368 (31 %)	Excellent	176 (15 %)

Mean CGPA 6.94 · attendance 78.5 % · coding score 57.3 · 30 % with an internship.
Injected data problems: 15 duplicates, ~2 % missing values in six columns, 6 impossible entries.

Limitations

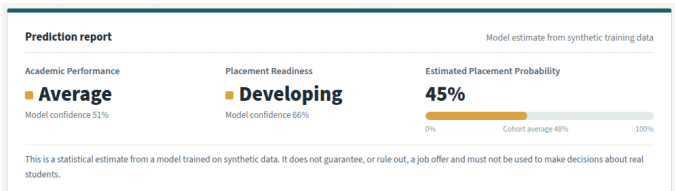
- The dataset is **synthetic**: results show that the pipeline recovers the simulated relationships, not that it predicts real placements.
- No prediction is a guarantee; outputs must not rank, filter or deny opportunities to real students.
- Age and gender are excluded from the models; outcomes are audited by group.

What the model relies on

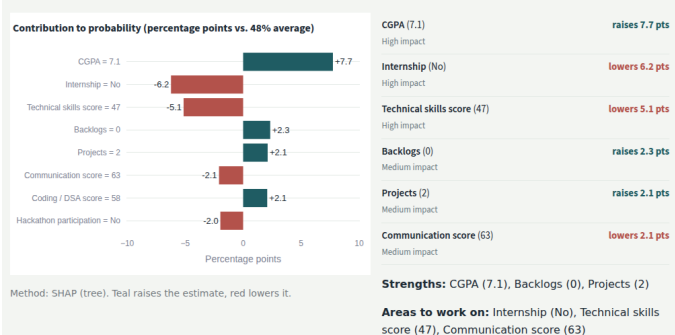


Permutation importance; the same four features lead the probability model.

Explained prediction in the application



What drove this estimate



SHAP contributions in percentage points for the “average student” sample profile.

Key observations

- No single algorithm won everywhere: a linear model fitted the near-linear readiness target best, boosted trees the non-linear probability target.
- Coding score, internship, CGPA and projects are the strongest placement drivers.
- An internship alone raises the average student's estimate from 45 % to 70 %.
- One explained prediction takes about 26.3 ms; the app validates all 16 inputs before any model runs.